

# Brian Kim

briankim31415@gmail.com | (214) 471-0577 | linkedin.com/in/briankim31415 | briankim31415.github.io

## EDUCATION

**MSE, Software Engineering & Systems** May 2025  
The University of Texas at Austin – Thesis: *Do Privacy Policies Align with Users' Privacy Values?*

**BSEE, Software Engineering & Design** May 2023  
The University of Texas at Austin – GPA: 3.73/4.0

## PROFESSIONAL EXPERIENCE

**Software Engineer**, UT Computational Visualization Center – Austin, TX May 2025 – May 2026

- Built a Python simulation platform around NVIDIA Sionna-RT to model ray-traced urban radio-wave propagation.
- Automated OpenStreetMap, ArcGIS, and Blender workflows to generate digital twin scenes in UE5 and Three.js.
- Orchestrated pipelines for real-time assessment including dataset generation and neural net training (ResNet-18).
- Trained a neural SDE Infinite-GAN on TACC supercomputer (400+ W&B GPU-hours) for urban movement generation.

**Graduate Research Assistant**, UT Center for Identity – Austin, TX September 2023 – May 2025

- Published 4 peer-reviewed papers under guidance of Dr. Suzanne Barber, including 2 first-authors.
- Served as a Teaching Assistant for 3 undergraduate and graduate courses in Information Security and Privacy.
- Assisted 2 senior capstone teams on the PrivacyCheck™ project with onboarding, planning, and development.

**Cybersecurity Intern**, United States Automobile Association – San Antonio, TX May 2023 – August 2023

- Migrated 8 legacy Detica rulesets to SAS, supporting the decommissioning of legacy fraud-detection systems.
- Partnered with onsite and offshore teams to refine matching logic and reduce false positives.

## AI/ML PROJECTS

**MAD-Community** October 2024 – December 2024

- Created novel Multi-Agent Debate framework utilizing LLM agents organized in interconnected communities.
- Designed directed graph network structure where communities debate and refine responses through multiple stages.
- Implemented system prompt engineering techniques to enhance debate quality and conciseness.
- Achieved 39% increase in accuracy over GPT-4 based model in original benchmark paper with optimized network design using significantly cheaper GPT-4o mini agents.

**NEAT-PSO: Hybrid CNN Architecture Evolution** October 2024 – December 2024

- Developed hybrid neural network architecture evolution technique by combining NeuroEvolution of Augmenting Topologies and Particle Swarm Optimization.
- Optimized NEAT hyperparameters using multi-objective PSO, targeting error reduction and parameter efficiency.
- Doubled accuracy with fittest individual over worst individual over 50 generations on CIFAR-10 image classification.

**Reinforcement Learning Trading Agent Exploration** March 2024 – April 2024

- Developed deep Reinforcement Learning model integrating CNNs with established frameworks for stock trading.
- Compared metrics such as Sharpe ratio, return, and volatility of RL algorithms A2C, PPO, DDPG, and their ensemble.
- Outperformed 4 major market indices using ensemble model and CNN over a 5-year period (Jan 2018 to Jan 2023).

## SKILLS

**Programming:** Python, Java, SAS, SQL, C++

**Machine Learning:** PyTorch, TensorFlow, scikit-learn, NumPy, Reinforcement Learning, OpenAI API, Codex

**Infrastructure:** Linux, Git, Slurm (HPC), Weights & Biases

**Simulation/Graphics:** Blender, Unreal Engine 5, PyGame, NVIDIA Sionna-RT, OpenStreetMap, ArcGIS